

G. NARAYANAMMA INSTITUTE OF TECHNOLOGY AND SCIENCE
(AUTONOMOUS)
Shaikpet, Hyderabad - 500 104

(For Women)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

IV B.Tech I Semester Academic Year 2026-27
Assignment-1

SUBJECT: DATA SCIENCE USING R

SUBJECT CODE: 127EY

Max Marks: 5

BRANCH: CSE-A, B, C and D

Blooms Taxonomy Levels:

Level 1 – Remembering, Level 2 – Understanding, Level 3 – Applying, Level 4 – Analyzing,
Level 5 – Evaluating, Level 6 - Creating

Answer all questions.

S.No.	Question	CO	BT Level
1 a)	Define Data Science and explain its role in modern data-driven decision-making.	CO2	L2
b)	List and explain the key stages of a Data Science project.	CO2	L2
2 a)	Analyze the common issues encountered while installing R and managing R packages. Explain their causes and recommend appropriate solutions.	CO1	L4
b)	Explain the major challenges involved in data collection and data management in a Data Science project.	CO1	L2
3 a)	Apply suitable R functions to explore and understand the structure and summary of a data frame using an example dataset.	CO1	L3
b)	Using an example dataset, apply appropriate R functions to detect missing values and outliers, and demonstrate suitable methods to handle them.	CO1	L3
4 a)	Explain the importance of Exploratory Data Analysis (EDA). Differentiate between univariate, bivariate, and multivariate data analysis with suitable examples.	CO1	L2
b)	What is data cleaning? Explain its importance in a Data Science project and discuss major data-cleaning techniques, including methods for handling missing values, duplicate records, inconsistent values, and outliers.	CO1	L2
5 a)	Evaluate different machine learning tasks, such as classification, regression, and clustering, for a given real-world problem and justify the most suitable task with appropriate reasons.	CO3	L5

b)	Evaluate the suitability of linear regression models for a given dataset. Discuss how model validation techniques influence the reliability of your results.	CO3	L-5
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